Mark Colley, FHEA

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RESEARCH INTERESTS

I develop computational methods that adapt interactive systems to individual users. The core method is human-in-the-loop multi-objective Bayesian optimization, which searches the design space of an interface using sparse and noisy ratings from participants. I complement this method with models of user behavior, including vision-language model personas as simulated participants and traffic-scale simulation in SUMO. The work thus connects to data science through Bayesian modeling of human feedback and to cognitive science through models of trust, attention, and workload. Laboratory, VR, and field studies evaluate the resulting designs on these constructs and on observed behavior. Automated vehicles and accessibility for blind, low vision, deaf, and hard of hearing people are the main application domains, because design errors there are safety critical and user needs vary widely. I publish data, code, and study materials openly, in line with FAIR principles.

EDUCATION

Ulm University Ulm, Germany 04/2019 — 26.04.2024
Doctor of Science (PhD) in Human–Computer Interaction
Thesis Title: *Calibrating Trust in Automated Vehicles-Theoretical, Design, & Empirical Insights into Effects of Visualizations on Trust*
Advisor: Prof. Dr. E. Rukzio
Committee: Prof. Dr. Stephen Brewster (University of Glasgow, UK), Prof. Dr. Wendy Ju (Cornell Tech, NYC, USA)

Cornell Tech New York, USA 01/2023 — 04/2023 & 07/2023 — 08/2023
Visiting Research Scholar with Prof. Dr. Wendy Ju
Note: supported by the German Academic Exchange Service (DAAD)

Ulm University Ulm, Germany 10/2015 — 11/2018
Master of Science in Computer Science Overall Grade: 1.1 - A-equivalent
Thesis Title: *Identification, Investigation and Classification of Use Cases for Cooperation in Highly Autonomous Driving and the Applicability thereof* Grade: 1.0 - A-equivalent

DHBW Ravensburg, Campus Friedrichshafen Friedrichshafen, Germany 09/2012 — 10/2015
Bachelor of Engineering: Computer Science
Note: cooperative degree as an employee of Airbus Defence and Space GmbH

JOB EXPERIENCE

Academia

University College London London, United Kingdom

Associate Professor 10/2026 — present
Lecturer/Assistant Professor 10/2024 — 09/2026
As the coordinator of the UCLIC Research Seminar Series, I organize and manage a vibrant seminar program featuring leading researchers in HCI. The recorded seminars are accessible through the UCLIC YouTube Channel. Through this, I also support the (re)presented initiative, supporting those who face travel limitations. I also lead two modules: PSYC0095 (Future Interfaces), which explores emerging technologies, & PSYC0097 (Interaction Design), focusing on design processes, theory, & practical application. As lab manager, I led the upgrade of all UCLIC labs, including the first strategic redesign of the central lab in more than a decade.

University of Tokyo Tokyo, Japan
Visiting Professor 04/2025 — 06/2025 & 01/2026 — 02/2026
Supported by Canon, I am a visiting Professor in the User Interface Research Group led by Takeo Igarashi.

Technische Hochschule Ingolstadt Ingolstadt, Germany
Visiting Professor 11/2025 — 11/2025
Supported by Technische Hochschule Ingolstadt, I am a visiting Professor in the Research Group led by Andreas Riener. I contributed to teaching in the modules *Design Principles 1*, *Data Visualization*, *AR/VR*, *Cognitive Evolution: The Impact of AI on Human Psychology*, and *Technology in HCI*.

TU Eindhoven Eindhoven, The Netherlands
Visiting Professor 12/2025 — 12/2025
Supported by Eindhoven Artificial Intelligence Systems Institute Visiting Professor Scholarship, I am a visiting Professor in the Research Group led by Pavlo Bazilinskyy.

Ulm University Ulm, Germany
Research Associate & Post-Doc 04/2019 — 09/2024
I significantly contributed to research and teaching and played a key role in securing and implementing three externally funded projects—INTUITIVER, SituWare, & SEMULIN. Furthermore, I successfully championed novel topics related to accessibility and computational methods into the department's research scope. My work incorporates numerous research approaches such as experiments,

interviews, focus groups, literature reviews, observation, usability testing, & dataset creation. It has led to over **40 peer-reviewed first-author full papers**. Between May and September 2024, I worked as a postdoctoral researcher.

Ulm University Ulm, Germany
Student Research Assistant 01/2019 — 03/2019
Led comprehensive literature reviews and significantly contributed to developing both a theoretical and an empirical publication on human-vehicle cooperation. My role involved critically analyzing existing works, synthesizing key findings, & collaboratively shaping the direction of the publication. Additionally, I actively participated in internal review processes.

——— Industry ———

Zefwih Neu-Ulm, Germany
Co-Founder & Research Lead 01/2022 — present
We are a startup specialising in Human–Computer Interaction, AI, XR, and immersive systems. Lead research-driven strategy and evaluation, overseeing user studies, prototyping, and human-centred design for industrial, cultural, and research partners. Bridge academic HCI expertise and applied product development to deliver accessible, usable, and innovative interactive systems and toolkits. Zefwih has won multiple national and international awards, including the Deutschland 4.0 Challenges with Gothaer ("inSURE") and WDR ("Öffy"), the Innovate2030 SDG9 programs with Bosch and Capgemini, and jury recognition at CHI Play, Deutscher Computerspielpreis, nextReality.Contest, and Games for Change.

Airbus Defence and Space GmbH Ulm, Germany
Software Engineer 09/2012 — 03/2019
I developed realistic simulations for aviation use cases, ensuring they closely mirror real-world conditions. My work included thorough testing to guarantee software reliability and performance, coupled with a comprehensive analysis of project requirements to create efficient and client-focused solutions. Additionally, I deployed and integrated these solutions, ensuring a smooth transition and optimal functionality within a safety-critical system that has existed and is being enhanced for >20 years. In 2014, I spent **3 months at Airbus Group in Newport, Wales**, contributing to system engineering tasks.

GRANTS

Received (Cash)

- Visiting Grant from the Technische Hochschule Ingolstadt as a Visiting Professor 2025 (2.000€)
- Recipient of the EAISI Visiting Professors Fellowship 2025 (7.500€)
- Recipient of the Canon Foundation Research Fellowship 2025 (7.500€)
- 2024 V&A Creative Experience Futures Lab (Co-Investigator, 10.000£)
- 2024 Cornell–UCL Global Strategic Collaboration Award: "Evaluating Cultural Differences in Mobility: Developing a Toolkit and Preliminary Investigations" (Co-Principal Investigator, 10.000$)

Total cash received: 36.000€

Received (In-Kind: Compute and Model Access)

- Anthropic AI for Science 2026 (sole PI; 50.000$ in API credits) — computational support for reproducibility and replication of research artifacts
- Google PaliGemma Academic Program (5.000$ in credits)

Total in-kind received: 55.000$

Member of the Advisory Board

- RapidMobility: LLM-based rapid prototyping for evaluating novel mobility concepts (PI: Patrick Ebel; value 60.000€)

Total amount: 60.000€

Named Cooperation Partner

- Computational Optimization of In-Vehicle User Interface Design for Calibrated Trust and Maximized Usability and Safety. **DFG** (Sachbeihilfe); applicant: Prof. Dr. E. Rukzio, Ulm University. Role: named international cooperation partner (WP1, WP3, WP4: implicit-feedback concept workshop, review of the longitudinal in-vehicle study design, guidelines and framework refinement). 36 months; 308.807€, thereof 2.000€ for two research stays in Ulm.

Total amount: 308.807€

MEMBERSHIPS, ACADEMIC AND INDUSTRIAL SUPPORT

- Fellow of the Higher Education Academy (FHEA), Advance HE (since 11/2025)
- Recipient of the Deutscher Akademischer Austauschdienst (DAAD, German Academic Exchange Service) research stipend 2022 for my research visit at Cornell Tech
- Startup your career support (10.000€) by Ulm University
- Hardware support by Leia Inc., NextMind (acquired by Snap), Nokia Bell Labs, and Meta
- Heidelberg Laureate Forum Foundation alumnus, AlumNode & HLFF Inspiring Minds member, mentor: Gabriele Kotsis
- Co-Founder of Zefwih since 01/2022, supported by the German Federal Ministry of Transport and Digital Infrastructure
- Affiliate member of the United Kingdom Advising and Tutoring (UKAT) association since 11/2024

SERVICE AND VOLUNTEERING ACTIVITIES

UCL Institutional Roles

- Member, UCLIC Staff Selection Committee, Lecturer in Human-AI Interaction (09/2026; 200+ applicants; all stages from short-listing to appointment)
- Coordinator, UCLIC Research Seminar Series (since 10/2024)
- Lab Manager, UCLIC (since 10/2024): upgrade of all UCLIC labs, including the first strategic redesign of the central lab in more than a decade
- Member, Global Disability Innovation Hub (GDI Hub) Lab Steering Committee (since 10/2026): lab space, operations, and collaboration with UCLIC
- Mentor, **UCL PaLS Mentoring Scheme** (Faculty of Brain Sciences), since 10/2026

Editorial Roles

- Senior Editor, **Behaviour & Information Technology** (since 02/2026; Associate Editor 12/2024 — 02/2026)
- Associate Editor, **IMWUT** (since 02/2024)
- Associate Editor, **IJHCS** (since 12/2024)
- Associate Editor, **IEEE Transactions on Human-Machine Systems** (since 08/2025)
- Member, ACM TOCHI Distinguished Reviewer Board (since 01/2026)

Conference Leadership and Organization

- Subcommittee Chair, Full Papers, AutoUI 2025
- **Best Paper Committee**, **CHI 2024 and 2026**
- AutoUI: Workshop Chair 2023, Demo Chair 2024, Workshop & Tutorial Chair 2025, Open Data Chair 2026
- 2026: UbiComp/ISWC Workshop Chair, ASSETS Workshop Chair, ACII Demo Chair
- MuC 2025, Workshop & Tutorial Chair
- **Main organizer**, Post-CHI Summer School on Automotive User Interfaces and Future Mobility (2023)
- Co-organizer, German Pre-CHI 2022, Ulm (more than 70 participants)

Program Committees

- Full papers: **CHI 2024 to 2027** (UX and Accessibility subcommittees), **CSCW 2024**, AutoUI 2021 to 2024, ETRA 2024 to 2027, MobileHCI 2023 to 2026, MuC 2022 to 2025, CUI 2024 and 2025, CHIWORK 2025, ISMAR 2025, NordiCHI 2026
- Late-Breaking Work: CHI 2022 to 2025, TEI 2023 and 2024, CHI Play 2024
- Student Research Track selection committee, AutoUI 2025; Juror, CHI 2026 Student Mentoring Program (Dissertation Research Roundtable)

Peer Review

- **Over 1000 peer reviews** for **Nature Computational Science**, **CHI**, **TVCG**, **CSCW**, **ToCHI**, **IJHCS**, IMWUT, UIST, MobileHCI, DIS, IUI, ISMAR, IEEE VR, VRST, AutoUI, ETRA, ICMI, EICS, CUI, TEI, CHI Play, TRF, IJHCI, ESWA; reviewer for the ERROR project
- Grant review: Member of the EPSRC Peer Review College (since 11/2024); NWO Open Competition Domain Science-M (2024); Linz Institute of Technology Seed Funding Program, JKU (2024)

Policy, Standards, and Examination

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- **Policy:** expert response to the Department for Transport (CCAV) consultation "Automated vehicles: statement of safety principles" (2026) on communicating vehicle intent (Principle 5) and safe interaction with vulnerable and disabled road users (Principle 7)
- Standards: former participant in the **ISO standardization process** for driving simulators
- External PhD examiner for Robin Connor Schramm, RheinMain University of Applied Sciences (2026)
- Examiner, Ulm University (since 11/2024): Media Informatics, Computer Science, Software Engineering, and AI; graded 10+ Master theses

Community Volunteering

- **Local council**, Erbach-Donaurieden, Germany (2019 — 2024)
- Former lifeguard, Deutsche Lebens-Rettungs-Gesellschaft e.V. (DLRG; German Life Saving Association)

MEDIA COVERAGE (Selection)

- Aerospace America (AIAA), "2025 simulations focus on emerging aircraft designs and crowded airspace" (January 2, 2026): mentions the Ingolstadt proof of concept added to UAM-SUMO.
- The Conversation, "How driverless vehicles can be made safer for deaf and hard of hearing people" (March 10, 2026).

TEACHING

Qualification: Fellow of the Higher Education Academy (FHEA) (Recognised 11/2025)

LOCCI: Living and Open Curriculum on Computational Interaction
Initial Contributor and Module Co-Author: Co-author, with Thomas Langerak, of the module *Human-in-the-Loop Optimization* (GitHub). The curriculum went live in October 2026 and is used at Aalto University, the Hasso Plattner Institute, and other institutions. At UCL, I use it in Interaction Design (since 2026/27) and Future Interfaces (from spring 2027). 2026 — present

Interaction Design (PSYC0097)
Course Organizer: Main organizer of the interdisciplinary course, in which students gain knowledge of methods for eliciting and specifying requirements, techniques for producing effective designs, prototyping methodology, and user-centred evaluation approaches. Fall 2025 — present

Future Interfaces (PSYC0095)
Course Organizer: Main organizer of the interdisciplinary course, emphasizing user-centered design and design thinking with an innovative approach to HCI. The course covered diverse topics, including AR/VR, personalization, Brain-computer interfaces, human-food interaction, and robotics. Spring 2025, 2026

Research Project in Human–Computer Interaction
Course Organizer: Co-organization of the interdisciplinary project, emphasizing user-centered design and design thinking, integrated with a year-long, research-driven group project that culminated in several publications. Fall 2019 — Fall 2023

Research Trends in Media Informatics
Course Organizer: Co-organization of the course, mentoring PhD students on course structure and content, & personally delivering in-depth, one-on-one instruction to over 15 students on conducting literature surveys using the PRISMA method, complemented by active involvement in student assessment and grading processes. Fall 2019 — Fall 2023

Automotive user interfaces and interactive vehicle applications
Course Organizer and Lecturer: Actively participated in the co-development of comprehensive course materials, aligning them with industry standards and academic requirements. Delivered an in-depth lecture focused on external communication of automated vehicles, a key component crucial for students' success in the final assessment. Contributed to the evaluation process by assisting in grading, ensuring a fair and thorough assessment of student performance. Fall 2022

PHD MENTORSHIP

I have the privilege of working with and mentoring fourteen PhD students and candidates as well as researchers.

Active

Ulm University
- Pascal Jansen (since 2021; submitted)
- Dr. Luca-Maxim Meinhardt (since 2022)
- Max Rädler (since 2024)
- Markus Sasalovici (since 2024)
- Julian Britten (since 2025)

TH Ingolstadt
- Mathias Haimerl (since 2021; submitted)

UCL
- Lan Xiao (since 2024)
- Coral Zawadzki (since 2025)
- Ramneek Ahluwalia (since 2025; primary supervisor)
- Catherine Kourou (since 2026; second supervisor, with Philip Treleaven; funded by Lloyds Bank)
- Rian Stephens (since 2026; primary supervisor, with Anthony Steed)

University of Tokyo
- Yotam Sechayk (since 2025)
- Dr. Xinyue Gui (since 2025)
- Ding Xia (since 2025)

Completed

Ulm University
- Dr. Annika Stampf (2021–2025)

VISITING RESEARCHER

2025
- Pascal Jansen (January - March)
- Carolin Stellmacher (June - September)
- Mengyang Ren (October - February'26)
- Marina Ricci (supported by Bando-STM (2500€); October - November)
- Leon Hanschmann (November - March'26)

2026
- Robin Connor Schramm (March - May, supported by DAAD)

THESIS SUPERVISION (Main Supervisor, Selection, Total>100)

Bachelor theses (all Ulm University unless noted)

- Simon Kopp (2024)
- Jonathan Westhauser (2023)
- Alexander Fassbender (2022)
- Julian Britten (2022; **now PhD student at Ulm University**)
- Elvedin Bajrovic (2021)
- Tim Fabian (2021)
- Max Rädler (2021)
- Svenja Krauß (2020)
- Jan Henry Belz (2020; **now PhD student at Ulm University in collab. with Porsche AG**)
- Stefanos Mytilineos (2020)
- Christian Bräuner (2019)

Master theses (multiple institutions)

- Ramneek Ahluwalia (2025; **now PhD student at UCL**)
- Shuai Yuai (2025)
- Marcel Giss (2025)
- Tim Fabian (2025)
- Julian Britten (2025; **now PhD student at Ulm University**)
- Svenja Krauß (2023)
- Max Rädler (2023; **now PhD student at Ulm University**)
- Bastian Wankmüller (2022)
- Cristina Evangelista (Ulm University and Cerence GmbH; 2022)
- Albin Zeqiri (2022; **now PhD student at Ulm University**)
- Annika Stampf (Ulm University and Mercedes Benz AG; 2021; **now PhD student at Ulm University**)
- Pascal Jansen (2021; **now PhD student at Ulm University**)
- Leon Dönch (University of Munich; 2020); co-supervised with Dr. Kai Holländer
- Gülsemin Dogru (Ulm University and Bosch GmbH; 2019)

INVITED TALKS, SEMINARS, AND COURSES (Selection)

Presentations automatically accompanying published conference papers are not listed.

Talks

- The Hong Kong Polytechnic University, Department of Industrial and Systems Engineering (15.04.2027): invited seminar, ISE seminar series; in-person
- S^3CIX 2026, "Theory in Computational Interaction" Workshop, University of Glasgow, UK (17.06.2026): "Bayesian Optimization as a Theory of Interaction"; invited theory pitch; in-person
- University of Tokyo Interactive Visual Intelligence Lab (24.02.2026): "Shaping the Future of Personalized Mobility"; in-person
- University of Tokyo User Interface Research Group (27.01.2026): "Approaches to studying cross-cultural dimensions of mobility"; in-person
- Shonan Meeting Building Trustworthy and Interactive Recommender Systems through Argumentation : "Evaluation and Impact of Argument-Based Recommendations"; in-person invited talk
- 2025 Huawei User Experience Design Technology Conference (UXTC) (24.11.2025): "HCI for mobility and beyond in the era of AI "; in-person
- World Usability Day 2025 (13.11.2025): "Shaping the Future of Mobility"; in-person
- Miraikan – The National Museum of Emerging Science and Innovation (16.05.2025); in-person
- University of Tokyo User Interface Research Group (09.05.2025): "Shaping the Future of Mobility"; in-person
- University of Cambridge Department of Engineering (31.03.2025); in-person
- Birmingham City University Research Centre (17.12.2024); in-person
- University of Sydney Design Lab (09.08.2024); in-person
- UC Calgary iLab (01.08.2023): "Automotive UI: The Intersection of Accessibility, Computational Methods, & Design", online
- KIT at Human-Centered Systems Lab (01.12.2022): "Inclusive Interaction with Future Mobility", in-person

Seminars

- Dagstuhl Seminar - Computational Theories of Interactive Behavior (Oct 19 – Oct 23, 2026)
- Shonan Meeting - Building Trustworthy and Interactive Recommender Systems through Argumentation (January 19-22, 2026)
- Decision & Design Spring Retreat (02.04.2025 - 04.04.2025; Guest speaker): "Shaping the Future of Mobility"; in-person
- Heidelberg Laureate Forum (23.09 - 30.09.2023)
- Dagstuhl Seminar - Radical Innovation and Design for Connected and Automated Vehicles (May 29 – Jun 03, 2022)

Courses / Tutorials

- 10th Summer School and Symposium on Computational Interaction (S^3CIX 2026), University of Glasgow, UK (17.06.2026): "Optimizing Interactive Mobility Interfaces with Human Feedback: Multi-Objective Bayesian Optimization in Practice"; invited course, 180-min session; in-person
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- Mensch und Computer 2025 (31.08.2025): "Human-in-the-Loop Bayesian Optimization for ALL: Practical Applications in Human-Computer Interaction Challenges"; tutorial with Pascal Jansen; in-person

AWARDS

UCL Open Science and Scholarship Award 2026, Winner in the category *Open-source software/analytical tools*: Awarded jointly by the UCL Office for Open Science and Scholarship and the UCL local network of the UK Reproducibility Network (UKRN). Selected from 50 applications and nominations across all categories.

ABSW Media Fellowship 2026: I am honored to have been selected as a 2026 Media Fellow by the Association of British Science Writers (ABSW), supported by UCL. This fellowship includes a placement at a major media outlet, offering a unique opportunity to bridge the gap between academic research and science journalism and to improve public engagement with complex HCI and mobility topics.

UCL Student Choice Awards 2026 (Nominated): Nominated by students in two categories: *Active Student Partnership* and *Brilliant Research-Based Education*. A total of 818 nominations were received across nine award categories for 479 individual staff members.

ACM SIGCHI Emerging Researcher Recognition 2025: I am honored to have been recognized by the ACM SIGCHI *for early-career contributions to inclusive HCI research, extensive community building, and championing open science in automotive user interfaces.*

UCL Computer Science Early Career Researcher of the Year 2025: I am honored to have been recognized by the Computer Science department of UCL as the *Early Career Researcher of the Year 2025.*

SaxFDM Open Data Award 2025: One of the winners for an open research data contribution demonstrating strong adherence to FAIR principles.

Open Science Award at AutoUI'26 - DOI: 10.1145/3828157.3828774
The award recognizes the public and FAIR release of all materials accompanying *Effects of Auditory Information for People With Visual Impairments in Highly Automated Vehicles*. Selection by the AutoUI 2026 award committee; the author had no role in the decision.

Outstanding Reviewer Recognition: AutoUI '21'25, CHI '21, CHI '22, 4xCHI '24, 2xCHI'25, CHI '26, IMWUT '23'26, MobileHCI '23'24 (3x), MuC '26, UIST'23'25'26, CHI Play '24 (3x), NordiCHI '26, IUI '27
I have been honored with twenty-six Outstanding Reviewer Awards and was named **Distinguished Reviewer** for MobileHCI'23, showing my commitment to excellence in academic review processes. These accolades reflect my deep understanding and critical analysis skills, contributing significantly to the advancement of scholarly discourse.

Honorable Mention Award at CHI'26 - DOI: 10.1145/3772318.3790882
We developed and tested a mobile Smart Pole Interaction Unit that communicates crossing cues alongside vehicle-mounted displays, and showed in a real-world field study that it improved clarity, trust, perceived safety, and reduced workload. The work contributes both a deployable interaction prototype and empirical evidence for how pedestrian-side interfaces can support safer and more legible automated-vehicle encounters.

Honorable Mention Award at CHI'26 - DOI: 10.1145/3772318.3790537
We investigated whether Vision-Language Model personas can support embodied HCI field studies by comparing 20 VLM personas with 20 human participants in an autonomous-vehicle street-crossing study. The work showed that VLM personas can approximate overall behavioral patterns and trust ratings, but miss human variability and nuance, making them useful for pilot testing, formative study design, and selective data augmentation rather than as a substitute for real participants.

Honorable Mention Award at CHI'26 - DOI: 10.1145/3772318.3790738
This paper investigates how external human-machine interfaces for automated vehicles can be made more inclusive for deaf and hard-of-hearing pedestrians. A formative focus group study identified clear state communication, multi-modal feedback, and combined visual cues as key design needs, which then informed a VR study comparing visual and speech-based eHMIs with hearing and DHH participants. The results show that visual eHMIs improved trust, perceived safety, usefulness, and crossing behaviour, speech improved subjective experience but not behaviour, and DHH participants looked at the vehicle longer, indicating distinct interaction needs that future eHMI design should address.

Honorable Mention Award at MUM'25 - DOI: 10.1145/3771882.3771888
AirClick is a space-saving system that transforms rooms on demand using modular, interactive inflatables. By connecting detachable modules to a floor-based air grid, users can instantly deploy furniture ranging from custom shelves to retail air mattresses. The system integrates seamlessly with traditional interiors, allowing users to rapidly reconfigure a space—such as turning a bedroom into an office—simply by clicking modules into place and activating them via touch or software.

Best Paper Award at WISS'25
To address the difficulties low-vision individuals face in maintaining context while using screen magnifiers, this study introduces *Graph Guide*, a semantic Focus+Context technique that dynamically projects out-of-view elements like axes and legends directly into the user's magnified view. Evaluation results indicate that *Graph Guide* significantly improves usability and reduces the navigational effort required for graph reading compared to standard magnification workflows and traditional overview maps.

Best Blue Sky Vision Paper Award at ICMI'25 (1.000$) - DOI: 10.1145/3716553.3750736
This paper presents Edmund's Journey, a 2035 design fiction that surfaces trade-offs among four AI superpowers, extended perception, cognitive offloading, externalized memory, and enhanced presence, and argues for an evaluation that preserves human capabilities and authenticity. It proposes a *Human Flourishing and Authenticity Benchmark* with four dimensions, cognitive preservation, autonomy and agency, skill development, and relational authenticity, using calibrated multiple-choice items, expert review, cross-cultural reporting, and a longitudinal component to assess sustained effects beyond accuracy.

Best Paper Award at AutoUI'25 - DOI: 10.1145/3744333.3747812
This paper introduces SPAT (Situational Prosocial and Aggressive Behavior in Traffic Scale). SPAT is a situational 12-item semantic differential scale that quantifies perceived prosocial and aggressive behavior in traffic for both human and automated road users, organized into three dimensions, Socialness, Awareness, and Predictability, and validated with exploratory and confirmatory factor analyses. In external validation on a test track, an automated vehicle that displayed deceleration intent was rated higher on socialness and awareness, and both deceleration and acceleration displays increased perceived predictability, showing sensitivity to interaction design.

Honorable Mention Award at CHI'25 - DOI: 10.1145/3706598.3713514
This paper addressed the challenge in scalable automotive user interface design. We implemented *OptiCarVis*, a system integrating Human-in-the-Loop Bayesian Optimization. An online study (N=117) demonstrates OptiCarVis efficacy in significantly improving trust, acceptance, perceived safety, and predictability without increasing cognitive load.

Honorable Mention Award at CHI'24 - DOI: 10.1145/3613904.3642341

This paper addressed the challenge in automotive user interface design, where testing transitions from lab-based driving simulators to on-road studies to increase ecological validity faces difficulties in replicating studies. A platform-portable infrastructure named *Portobello* facilitates running identical studies in-lab and on-road. A proof-of-concept was demonstrated by integrating Portobello with the on-road simulator XR-OOM with 32 participants in both settings.

Honorable Mention Award at MobileHCI'23 - DOI: 10.1145/3604275
We delved into the impact of Infinite Scrolling on social media usage, particularly focusing on how it contributes to habitual and regretful use. Our study (N=46) defined and examined the 'loop' phenomenon that traps users in extended sessions and also emphasized the importance of considering the user's context in designing interventions, revealing that factors outside the app itself often influence breaks in social media usage.

Best Video Award at AutoUI'23 - DOI: 10.1145/3581961.3609854
I was honored to receive the Best Video Award, chosen by popular vote, for a video presentation in which we explored the nuanced distinctions between autonomous, automated, & highly automated vehicles. The video aimed to stimulate a balanced discussion on the potential impacts—positive, negative, & ambiguous—of truly autonomous vehicles that can act independently, possibly even against the wishes of their owners.

Honorable Mention Award at MobileHCI'22 - DOI: 10.1145/3546712
I was honored to receive an Honorable Mention Award for a full paper that developed a taxonomy of augmented reality visualizations for connected automated and manual driving, aiming to enhance trust in such systems. Our work, which included an evaluative driving simulator study, focused on how augmented reality can help drivers and passengers understand and trust the information provided by infrastructure-mounted sensors and onboard systems.

Honorable Mention Award at CHI'20 - DOI: 10.1145/3313831.3376472
We presented an inclusive user-centered design for vehicle-pedestrian communication to enhance the safety and experience of both vision-impaired and sighted pedestrians. Workshops with vision-impaired individuals and a virtual reality study revealed that communication from all relevant vehicles and detailed messaging significantly improve trust and understanding and reduce cognitive load.

SOFTWARE (Selection)

colleyRstats (CRAN package; DOI: 10.32614/CRAN.package.colleyRstats)
Functions to streamline reproducible statistical analysis, visualization, and APA-style reporting in R. Used by more than 100 students.
CRAN — GitHub

Bayesian Optimization for Unity (with Pascal Jansen)
Human-in-the-loop multi-objective Bayesian optimization of interface designs in Unity, built on BoTorch. The approach is used at IT:U Linz and the University of Tokyo.
GitHub — MuC 2025 tutorial

botorch-rust (PyPI package)
Unofficial Rust implementation of the BoTorch core (exact Gaussian processes, acquisition functions, acquisition optimization) with BoTorch's Python API, validated against BoTorch. A 30-iteration Bayesian optimization loop runs in 351 ms instead of 16.6 s, without a PyTorch dependency.
PyPI — GitHub

reprobe
Docker-sandboxed reproducibility harness for open-data artifacts. It takes a repository URL and returns a runnable check with an ACM and FAIR badge report. Used for the AutoUI 2026 artifact checks.
GitHub

mechcheck
Mechanical checks for LaTeX papers and theses with 117 rules, including bibliography verification against Crossref, OpenAlex, and DBLP, accessibility, anonymity, and submission rules for CHI, ASSETS, AutomotiveUI, IMWUT, and Transportation Research Part F. Runs in the browser, inside Overleaf, or in CI.
GitHub

PedSUMO, UAM-SUMO, HUD-SUMO (HRI 2024, HRI 2025)
Extensions of the SUMO traffic simulator for automated vehicle and pedestrian interaction, urban air mobility, and in-vehicle head-up displays. They estimate population-level effects of interface designs. Aerospace America (AIAA) featured UAM-SUMO in its 2025 simulation review.
PedSUMO — UAM-SUMO — HUD-SUMO

citysample-carla (Zenodo; DOI: 10.5281/zenodo.22306770)
Runs Epic's City Sample as a CARLA driving simulator with the real city geometry, CARLA traffic, and CARLA sensors. The extracted road network covers 3,337 roads, 172 junctions, and 145 km, with traffic lights, pedestrian navigation, and semantic segmentation.
GitHub

VIP-Sim (UIST 2025; with Max Rädler and Enrico Rukzio)
Gaze-contingent overlay that simulates 21 vision impairment symptoms on any Windows or macOS application, developed with people with vision impairments. Version 2.0 adds working macOS builds, plain-language symptom names, and keyboard and high-contrast operation of the tool itself. Used in classrooms at several universities, including the University of Bremen.
GitHub — Paper

UnitEye (MuC 2024; joint first author with Tobias Wagner)
Webcam-based eye-tracking plugin for Unity with per-user calibration, area-of-interest tracking, and gaze logging. Runs on Windows, macOS, Linux, and WebGL. VIP-Sim uses it for gaze-contingent effects.
GitHub — Paper

DAGitty, accessibility-focused fork
Beginner-friendly version of the DAGitty causal-diagram editor by Johannes Textor and Benito van der Zander, with export to R (dagitty, ggdag), LaTeX, and image formats.
Live version — GitHub

MyoSuite (named contributor to the MyoSuite 3 release)
Musculoskeletal environments and tasks simulated with the MuJoCo physics engine and wrapped in the Gym API, for machine learning on biomechanical control problems.
GitHub — myosuite.org

SKILLS

- Highly proficient in Leadership. Based on over 6 years in mentoring and leading Bachelor's, Master's, and PhD students, I am a highly effective and appreciated leader. In the academic year 25/26, I also advanced my skills through the competitive *Advancing PI Leadership Programme* at UCL.
- Coding: Programming in **R** (data analysis, visualization, web-scraping), Python (application of neural networks), Java (professionally at Airbus), & C# (especially with **Unity**)
- Research: **Quantitative Analysis** using R; experience with parametric and non-parametric data; linear regression and hierarchical models
- Research: **Qualitative Analysis** according to Saldaña
- Proficient in open science practices, encompassing transparency, reproducibility, & accessibility in research; experienced in sharing data (e.g, here) via open access platforms, utilizing open-source tools, contributing to community projects, & advocating for ethical research aligned with FAIR principles.
- Languages: German (native), English (proficient), French, Spanish (Intermediate), Latin

MAJOR PUBLICATIONS (CHI, IMWUT, UIST)

ACM CHI, IMWUT, and UIST are widely recognized as the premier venues for publishing research in the field of HCI. They are highly competitive, with acceptance rates typically ranging between 20-25%.

CHI

1. X. Gui, D. Xia, **M. Colley**, Y. Li, V. Chauhan, A. Anubhav, Z. Zhou, E. Javanmardi, S. H. Seo, C.-M. Chang, M. Tsukada, & T. Igarashi, Peeking Ahead of the Field Study: Exploring VLM Personas as Support Tools for Embodied Studies in HCI
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790537
   CHI Honorable Mention Award for Best Paper (top 5%); † corresponding author
2. V. Chauhan, A. Anubhav, **M. Colley**, C.-M. Chang, X. Gui, D. Xia, E. Javanmardi, T. Igarashi, K. Fujiwara, & M. Tsukada, Don't Worry, Just Follow Me: Prototyping and In-the-Wild Evaluation of Smart Pole Interaction Unit with Mobility
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790882
   CHI Honorable Mention Award for Best Paper (top 5%); † corresponding author
3. P. Jansen*, J. Britten*, **M. Colley***, M. Sasalovici, & E. Rukzio, MIRAGE: Enabling Real-Time Automotive Mediated Reality
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3791195, *Joint First Authors
4. S. Lämmer, **M. Colley**, & P. Ebel, GTA: Generative Traffic Agents for Simulating Realistic Mobility Behavior
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790772
5. C. Stellmacher, L. Dratzidis, A. Zenner, I. Wald, J. Schöning, Y. Rogers, D. Degraen, & **M. Colley**, Understanding How Mobile Interactions Shape Grasp and Contact Patterns Beyond the Touchscreen
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790565
6. J. Susak, Y. Liu, P. Jansen, & **M. Colley**, ProVoice: Designing Proactive Functionality for In-Vehicle Conversational Assistants using Multi-Objective Bayesian Optimization to Enhance Driver Experience
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3791877
7. Y. Sechayk, H. Rave, M. Rädler, **M. Colley**, Z. Zhou, A. Shamir, & T. Igarashi, Improving Low-Vision Chart Accessibility via On-Cursor Visual Context
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3791165
8. **M. Colley**, S. Kopp, D. Dey, P. Jansen & E. Rukzio, eHMI for All - Investigating the Effect of External Communication of Automated Vehicles on Pedestrians, Manual Drivers, and Cyclists in Virtual Reality
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790585
9. W. Xu, F. Hajiseyedjavadi, T. T. M. Tran, & **M. Colley**, Exploring the Impacts of Background Noise on Auditory Stimuli of Audio-Visual eHMIs for Hearing, Deaf, and Hard-of-Hearing People
   In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3791557
10. W. Xu, F. Hajiseyedjavadi, K. Weir, C. Eze, & **M. Colley**, Towards Inclusive External Human-Machine Interface: Exploring the Effects of Visual and Auditory eHMI for Deaf and Hard-of-Hearing People
    In Proc. of CHI 2026, ACM, doi: 10.1145/3772318.3790738
    CHI Honorable Mention Award for Best Paper (top 5%)
11. P. Jansen*, **M. Colley***, S. Krauß, D. Hirschle, & E. Rukzio, OptiCarVis: Improving Automated Vehicle Functionality Visualizations Using Bayesian Optimization to Enhance User Experience
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713514, *Joint First Authors
    CHI Honorable Mention Award for Best Paper (top 5%)
12. **M. Colley***, P. Jansen*, M. Keskar, & E. Rukzio, Improving External Communication of Automated Vehicles Using Bayesian Optimization
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3714187, *Joint First Authors
13. **M. Colley**, J. Westhauser, J. Andersson, A. G. Mirnig, & E. Rukzio, Introducing ROADS: A Systematic Comparison of Remote Control Interaction Concepts for Automated Vehicles at Road Works
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713476
14. M. Sasalovici, A. Zeqiri, R. C. Schramm, O. J. A. Nuñez, P. Jansen, J.P. Freiwald, **M. Colley**, C. Winkler, & E. Rukzio, Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3714077
15. L. Meinhardt, L. Wilke, M. Elhaidary, J. von Abel, P. Fink, M. Rietzler, **M. Colley**, & E. Rukzio, Light My Way: Developing and Exploring a Multimodal Interface to Assist People With Visual Impairments to Exit Highly Automated Vehicles
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713454
16. L. Meinhardt, C. Schramm, P. Jansen, **M. Colley**, & E. Rukzio, Fly Away: Evaluating the Impact of Motion Fidelity on Optimized User Interface Design via Bayesian Optimization in Automated Urban Air Mobility Simulations
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713288
17. L. Meinhardt, M. Elhaidary, **M. Colley**, M. Rietzler, J. O. Rixen, A. K. Purohit, & E. Rukzio, Scrolling in the Deep: Analysing Contextual Influences on Intervention Effectiveness during Infinite Scrolling on Social Media
    In Proc. of CHI 2025, ACM, doi: 10.1145/3706598.3713187
18. **M. Colley**, O. Rajabi, & E. Rukzio, Investigating the Effects of External Communication and Platoon Behavior on Manual Drivers at Highway Access
    In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642365
19. **M. Colley**, B. Wanner, M. Rädler, M. Rötzer, J. Frommel, T. Hirzle, P. Jansen, & E. Rukzio, Effects of a Gaze-Based 2D Platform Game on User Enjoyment, Perceived Competence, & Digital Eye Strain
    In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3641909
20. D. Dey, T. Senan, B. Hengeveld, **M. Colley**, A. Habibovic, & W. Ju, Multi-Modal eHMIs: The Relative Impact of Light and Sound in AV-Pedestrian Interaction
    In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642031
21. F. Bu, S. Li, D. Goedicke, **M. Colley**, G. Sharma, & W. Ju, Portobello: Extending Driving Simulation from the Lab to the Road
    In Proc. of CHI 2024, ACM, doi: 10.1145/3613904.3642341
    CHI Honorable Mention Award for Best Paper (top 5%)
22. P. Jansen, J. Britten, A. Häusele, T. Segschneider, **M. Colley**, & E. Rukzio, AutoVis: Enabling Mixed-Immersive Analysis of Automotive User Interface Interaction Studies
    In Proc. of CHI 2023, ACM, doi: 10.1145/3544548.3580760, [Website Link]
23. M. Lanzer, I. Koniakowsky, **M. Colley** and M. Baumann, Interaction Effects of Pedestrian Behavior, Smartphone Distraction and External Communication of Automated Vehicles on Crossing and Gaze Behavior
    In Proc. of CHI 2023, ACM, doi: 10.1145/3544548.3581303
24. J. O. Rixen, **M. Colley**, A. Askari, J. Gugenheimer, & E. Rukzio, Consent in the Age of AR: Investigating The Comfort With Displaying Personal Information in Augmented Reality
    In Proc. CHI 2022, ACM, doi: 10.1145/3491102.3502140
25. **M. Colley**, E. Bajrovic, & E. Rukzio, Effects of Pedestrian Behavior, Time Pressure, & Repeated Exposure on Crossing Decisions in Front of Automated Vehicles Equipped with External Communication
    In Proc. CHI 2022, ACM, doi: 10.1145/3491102.3517571
26. J. O. Rixen, T. Hirzle, **M. Colley**, Y. Etzel, E. Rukzio and J. Gugenheimer, Exploring Augmented Visual Alterations in Interpersonal Communication

In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445597
27. **M. Colley**, B. Eder, J. O. Rixen, & E. Rukzio, Effects of Semantic Segmentation Visualization on Trust, Situation Awareness, & Cognitive Load in Highly Automated Vehicles
In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445351
28. K. Holländer*, **M. Colley***, E. Rukzio and A. Butz, A Taxonomy of Vulnerable Road Users for HCI Based On A Systematic Literature Review
In Proc. CHI 2021, ACM, doi: 10.1145/3411764.3445480
29. **M. Colley**, M. Walch, J. Gugenheimer, A. Askari, & E. Rukzio, Towards Inclusive External Communication of Autonomous Vehicles for Pedestrians with Vision Impairments
In Proc. CHI 2020, ACM, doi: 10.1145/3313831.3376472
CHI Honorable Mention Award for Best Paper (top 5%)
30. K. Rogers, **M. Colley**, D. Lehr, J. Frommel, M. Walch, L. E. Nacke and M. Weber, KickAR: Exploring Game Balancing Through Boosts and Handicaps in Augmented Reality Table Football
In Proc. CHI 2018, ACM, doi: 10.1145/3173574.3173740

IMWUT

1. **M. Colley***, S. Hartwig*, A. Zeqiri, T. Ropinski, & E. Rukzio, AutoTherm: A Dataset and Benchmark for Thermal Comfort Estimation Indoors and in Vehicles
In Proc. IMWUT 2024, ACM, *Joint First Authors, doi: 10.1145/3678503
2. A. Stampf, **M. Colley**, A. Knuth, C. Tasci, & E. Rukzio, Examining Psychological Conflict-Handling Strategies for Highly Automated Vehicles to Resolve Legal User-Vehicle Conflicts
In Proc. IMWUT 2024, ACM, doi: 10.1145/3678511
3. A. Stampf, M. Sasalovici, L. Meinhardt, **M. Colley**, M. Giss, & E. Rukzio, Move, Connect, Interact: Introducing a Design Space for Cross-Traffic Interaction
In Proc. IMWUT 2024, ACM, doi: 10.1145/3678580
4. L. Meinhardt, M. Rück, J. Zähnle, M. Elhaidary, **M. Colley**, M. Rietzler, & E. Rukzio, Hey, What's Going On? Conveying Traffic Information to People with Visual Impairments in Highly Automated Vehicles: Introducing OnBoard
In Proc. IMWUT 2024, ACM, doi: 10.1145/3659618
5. **M. Colley**, O. Speidel, J. Strohbeck, J. O. Rixen, J. H. Belz, & E. Rukzio, Effects of Uncertain Trajectory Prediction Visualization in Highly Automated Vehicles on Trust, Situation Awareness, & Cognitive Load
In Proc. IMWUT 2023, ACM, doi: 10.1145/3631408, [Video Link]
6. **M. Colley***, L. Meinhardt*, A. Fassbender, M. Rietzler, & E. Rukzio, Come Fly With Me - Investigating the Effects of Path Visualizations in Automated Urban Air Mobility
In Proc. IMWUT 2023, ACM, doi: 10.1145/3596249, *Joint First Authors
7. **M. Colley**, J. Britten, & E. Rukzio, Scalability in External Communication of Automated Vehicles: Evaluation and Recommendations
In Proc. IMWUT 2023, ACM, doi: 10.1145/3596248
8. P. Jansen, **M. Colley**, & E. Rukzio, A Design Space for Human Sensor and Actuator Focused In-Vehicle Interaction Based on a Systematic Literature Review
In Proc. IMWUT 2022, ACM, doi: 10.1145/3534617
9. **M. Colley**, M. Rädler, J. Glimmann, & E. Rukzio, Effects of Scene Detection, Scene Prediction, & Maneuver Planning Visualizations on Trust, Situation Awareness, & Cognitive Load in Highly Automated Vehicles
In Proc. IMWUT 2022, ACM, doi: 10.1145/3534609, [Video Link]
10. **M. Colley**, P. Jansen, E. Rukzio and J. Gugenheimer, SwiVR-Car-Seat: Exploring Vehicle Motion Effects on Interaction Quality in Virtual Reality Automated Driving Using a Motorized Swivel Seat
In Proc. IMWUT 2021, ACM, doi: 10.1145/3494968
11. **M. Colley**, S. Krauß, M. Lanzer, & E. Rukzio, How Should Automated Vehicles Communicate Critical Situations? A Comparative Analysis of Visualization Concepts
In Proc. IMWUT 2021, ACM, doi: 10.1145/3478111

UIST

1. M. Rädler, J. Zähnle, L.-M. Meinhardt, P. Jansen, E. Rukzio, and **M. Colley**. 2026. MoTUI: Personalization of In-Vehicle Tactile Interfaces for People With Vision Impairments and the Blind
In Proc. UIST 2026, ACM, doi: *accepted*
2. X. Gui, D. Xia, Y. Li, **M. Colley**, C. M. Chang, S. Seo, M. Tsukada & T. Igarashi, Tap2Stop: Pedestrian-Activated Emergency Stop for Autonomous Vehicles
In Proc. UIST 2026, ACM, doi: *accepted*
3. M. Rädler, **M. Colley**, & E. Rukzio, VIP-Sim: A User-Centered Approach to Vision Impairment Simulation for Accessible Designs
In Proc. UIST 2025, ACM, doi: 10.1145/3746059.3747704

FURTHER PUBLICATIONS

Journal paper
Transportation Research Part F, with an impact factor of 4.60 (2022), and the International Journal of Human-Computer Studies, with an impact factor of 5.40 (2024), are considered to be top-tier journals in traffic psychology.

1. P. Jansen*, **M. Colley***, M. Rädler, J. Schwedler, & E. Rukzio, Longitudinal effects of visualizing uncertainty of situation detection and prediction of automated vehicles on user perceptions
In Transportation Research Part F: Traffic Psychology and Behaviour 2025, Elsevier, *Joint First Authors, doi: 10.1016/j.trf.2025.05.013
2. P. Jansen*, **M. Colley***, T. Pfeifer, & E. Rukzio, Visualizing Imperfect Situation Detection and Prediction in Automated Vehicles: Understanding Users
In Transportation Research Part F: Traffic Psychology and Behaviour 2024, Elsevier, *Joint First Authors, doi: 10.1016/j.trf.2024.05.015
3. A. Stampf, A. Knuth, **M. Colley**, & E. Rukzio, Law and Order: Investigating the Effects of Conflictual Situations in Manual and Automated Driving in a German Sample
In International Journal of Human-Computer Studies 2024, Elsevier, doi: 10.1016/j.ijhcs.2024.103260
4. **M. Colley**, B. Wankmüller, T. Mend, T. Väth, E. Rukzio and J. Gugenheimer, User Gesticulation Inside an Autonomous Vehicle with External Communication can Cause Confusion in Pedestrians and a Lower Willingness to Cross
In Transportation Research Part F: Traffic Psychology and Behaviour 2022, Elsevier, doi: 10.1016/j.trf.2022.03.011
5. **M. Colley**, C. Hummler, & E. Rukzio, Effects of Mode Distinction, User Visibility, & Vehicle Appearance on Mode Confusion When Interacting With Highly Automated Vehicles
In Transportation Research Part F: Traffic Psychology and Behaviour 2022, Elsevier, doi: 10.1016/j.trf.2022.06.020
6. **M. Colley**, S. Mytilineos, M. Walch, E. Rukzio, & J. Gugenheimer, Requirements for the Interaction With Highly Automated Construction Site Delivery Trucks
Frontiers, 2022, doi: 10.3389/fhumd.2022.794890

7. M. Lanzer, T. Stoll, **M. Colley**, & M. Baumann, Intelligent Mobility in the City: The Influence of System and Context Factors on Drivers' Takeover Willingness and Trust in Automated Vehicles
, Frontiers, 2021, doi: 10.3389/fhumd.2021.676667

Conference full paper

1. Y. Zhang, Z. Lai, J. Wang, C. Yu, J. Chang, T. Olugbade, M. Nygren, B. Petreca, Y. Cho, S. Baurley, X. Wei, **M. Colley**, & N. Bianchi-Berthouze, *ComfyFashionTryOn: A Multimodal Dataset for Try-On Clothing Comfort Detection in Women. In Proc. 14th International Conference on Affective Computing and Intelligent Interaction (ACII 2026).* **Oral Presentation**.
2. **Colley, M.***, Jansen, P.*, Krauß, S., Rukzio, E. Multi-Session User Experience Assessments of Computationally Optimized Automated Vehicle Functionality Visualizations.
In Proc. AutoUI'26, ACM, doi: https://doi.org/10.1145/3828157.3828785; * Joint First Authors
3. **Colley, M.**, Aescht, T., Rajabi, O., Rädler, M., Jansen, P., Rukzio, E. Effects of Auditory Information for People With Visual Impairments in Highly Automated Vehicles.
In Proc. AutoUI'26, ACM, doi: https://doi.org/10.1145/3828157.3828774
AutomotiveUI Open Science Award
4. Fink, P. D. S., Brown, J. R., Niane, H., Movalia, R. M., James, K. J., Howe, A., **Colley, M.,** Giudice, N. A. I Can See Clearly Now: Co-Designing Augmented Reality Supports for Safer Pedestrian Detection by Older Adult Drivers
In Proc. AutoUI'26 (accepted)
5. Y. Sechayk and Y. Li and H. Rave and A. Shamir and **M. Colley** and T. Igarashi, Graph Guide:低視力者支援のためのセマンティック Focus+Context グラフ表示 (translation: Graph Guide: Semantic Focus+Context Graph Display for Low Vision Support)
In Proc. WISS 2025
WISS Best Paper Award
6. P. Jansen and B. Hölz and J. Britten and **M. Colley** and E. Rukzio, AirClick: Modularized Interactive Inflatables for On-Demand Room Transformation
In Proc. MUM 2025, ACM, doi: 10.1145/3771882.3771888
MUM Honorable Mention Award
7. D. Xia and X. Gui and F. Gao and D. Li and **M. Colley** and T. Igarashi, Automating eHMI Action Design with LLMs for Automated Vehicle Communication
In Findings of EMNLP 2025, ACL Anthology, doi: 10.18653/v1/2025.findings-emnlp.148
8. Y. Li and X. Gui and D. Xia and **M. Colley** and T. Igarashi, TailCue: Exploring Animal-inspired Robotic Tail for Automated Vehicles Interaction
In Proc. HAI 2025, ACM, doi: 10.1145/3765766.3765767; † corresponding author
9. S. Zepf and **M. Colley**, Human Authenticity and Flourishing in an AI-Driven World: Edmund's Journey and the Call for Mindfulness
In Proc. ICMI 2025, ACM, doi: 10.1145/3716553.3750736
ICMI Best Blue Sky Vision Paper Award
10. H. İppoliti, **M. Colley**, D. Dey, P. Wintersberger, S. Sadeghian, A. Löcken, A. Matviienko, A. Habibovic, H. Müller, A. Hildebrandt, and S. Boll. SPAT: Situational Prosocial and Aggressive Behavior Perception in Traffic Scale
In Proc. AutoUI 2025, ACM, doi: 10.1145/3744333.3747812
AutomotiveUI Best Paper Award
11. K. Terao, I. Mandel, M. Franchi, C. Yang, **M. Colley**, and W. Ju. Evaluating Interfaces for Non-Driving Related Tasks While Operating an E-Scooter
In Proc. AutoUI 2025, 10.1145/3744333.3747808
12. L. Xiao, M. Bandukda, F. Mingzhe Li, **M. Colley**, and C. Holloway. Understanding the Video Content Creation Journey of Creators with Sensory Impairment in Kenya
In Proc. ASSETS 2025, 10.1145/3663547.3746356
13. L. Meinhardt, S. Demharter, M. Rietzler, **M. Colley**, T. Eßmeyer, and E. Rukzio. Mind Games! Exploring the Impact of Dark Patterns in Mixed Reality Scenarios
In Proc. MobileHCI 2025, 10.1145/3743709
14. P. Fink, H. Milne, A. Caccese, M. Alsamsam, J. Loranger, **M. Colley**, N. Giudice, Accessible Maps for the Future of Inclusive Ridesharing
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675736
15. A. Asha, **M. Colley**, S. Sultana, E. Sharlin, Introducing AV-Sketch: An Immersive Participatory Design Tool for Automated Vehicle — Passenger Interaction
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675705
16. P. Ebel, P. Bazilinskyy, **M. Colley**, C. Goodridge, P. Hock, C. Janssen, H. Sandhaus, A. Srinivasanm, P. Wintersberger, Changing Lanes Toward Open Science: Openness and Transparency in Automotive User Research
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675730
17. A. Stampf, **M. Colley**, B. Girst, E. Rukzio, Exploring Passenger-Automated Vehicle Negotiation Utilizing Large Language Models for Natural Interaction
In Proc. AutoUI 2024, ACM, doi: 10.1145/3640792.3675725
18. A. Stampf* and **M. Colley***, Deriving Non-Driving-Related Activities in Highly Automated Driving via an Autoethnographic Approach by Traveling Canada in a Recreational Vehicle
In Proc. MuC 2024, *Joint First Authors, doi: 10.1145/3670653.3670663
19. T. Wagner*, **M. Colley***, D. Breckel, M. Kösel, & E. Rukzio, UnitEye: Introducing a User-Friendly Plugin to Democratize Eye Tracking Technology in Unity Environments
In Proc. MuC 2024, *Joint First Authors, doi: 10.1145/3670653.3670655,
20. **M. Colley***, P. Jansen*, J. J. Matthiesen*, H. Hoberg, C. Reger, & I. Thiermann, How Much Home Office is Ideal? A Multi-Perspective Algorithm
In Proc. CHIWORK 2023, ACM, doi: 10.1145/3596671.3596672, *Joint First Authors
21. **M. Colley**, A. Stampf, W. Fischer, & E. Rukzio, Effects of 3D Displays on Mental Workload, Situation Awareness, Trust, & Performance Assessment in Automated Vehicles
In Proc. MUM 2023, ACM, doi: 10.1145/3626705.3627786
22. **M. Colley**, C. Evangelista, T. D. Rubiano, & E. Rukzio, Effects of Urgency and Cognitive Load on Interaction in Highly Automated Vehicles
In Proc. MobileHCI 2023, ACM, doi: 10.1145/3604254
23. J. O. Rixen, L. Meinhardt, M. Glöckler, A. Schlothauer, M. Ziegenbein, **M. Colley**, J. Gugenheimer, & E. Rukzio, The Loop and Reasons to Break It: Investigating Infinite Scrolling Behaviour in Social Media Applications and Reasons to Stop
In Proc. MobileHCI 2023, ACM, doi: 10.1145/3604275
MobileHCI Honorable Mention Award for Best Paper (top 5%)
24. S. Suzuki, **M. Colley**, S. Li, I. Mandel, A. Stampf and W. Ju, AdVANcing Design: Customizing Spaces for Vanlife
In Proc. AutoUI 2023, ACM, doi: 10.1145/3580585.3607175
25. M. Woide, L. Miller, **M. Colley**, N. Damm and M. Baumann, I've Got the Power: Exploring the Impact of Cooperative Systems on Driver-Initiated Takeovers and Trust in Automated Vehicles
In Proc. AutoUI 2023, ACM, doi: 10.1145/3580585.3607165
26. **M. Colley**, J. O. Rixen, W. I. Pellegrino, & E. Rukzio, (Eco-)Logical to Compare? - Utilizing Peer Comparison to Encourage Ecological

Driving in Manual and Automated Driving
In Proc. AutoUI 2022, ACM, doi: 10.1145/3543174.3545256
27. M. Woide, **M. Colley**, N. Damm, & M. Baumann, Effect of System Capability Verification on Conflict, Trust, & Behavior in Automated Vehicles
In Proc. AutoUI 2022, ACM, doi: 10.1145/3543174.3545253
28. A. Stampf, **M. Colley**, & E. Rukzio, Towards Implicit Interaction in Highly Automated Vehicles - A Systematic Literature Review
In Proc. MobileHCI 2022, ACM, doi: 10.1145/3546726
29. **M. Colley**, T. Fabian, & E. Rukzio, Investigating the Effects of External Communication and Automation Behavior on Manual Drivers at Intersections
In Proc. AutoUI 2022, ACM, doi: 10.1145/3546711
30. P. Hock*, **M. Colley***, A. Askari, T. Wagner, M. Baumann, & E. Rukzio, Introducing VAMPIRE – Using Kinaesthetic Feedback in Virtual Reality for Automated Driving Experiments
In Proc. AutoUI 2022, ACM, doi: 10.1145/3543174.3545252, * Joint First Authors
31. **M. Colley**, J. Britten, S. Demharter, T. Hisir, & E. Rukzio, Feedback Strategies for Crowded Intersections in Automated Traffic — A Desirable Future?
In Proc. AutoUI 2022, ACM, doi: 10.1145/3543174.3545255
32. M. Haimerl, **M. Colley** and A. Riener, Evaluation of Common External Communication Concepts of Automated Vehicles for People With Intellectual Disabilities
In Proc. MobileHCI 2022, ACM, doi: 10.1145/3546717
33. T. Müller*, **M. Colley***, G. Dogru, & E. Rukzio, AR4CAD: Creation and Exploration of a Taxonomy of Augmented Reality Visualization for Connected Automated Driving
In Proc. MobileHCI 2022, ACM, doi: 10.1145/3546712
MobileHCI Honorable Mention Award for Best Paper (top 5%)
34. **M. Colley**, B. Wankmüller, & E. Rukzio, A Systematic Evaluation of Solutions for the Final 100m Challenge of Highly Automated Vehicles
In Proc. MobileHCI 2022, ACM, doi: 10.1145/3546713
35. **M. Colley**, J. H. Belz, & E. Rukzio, Investigating the Effects of Feedback Communication of Autonomous Vehicles
In Proc. AutoUI 2021, ACM, doi: 10.1145/3409118.3475133
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1. **M. Colley**, J. Czymmeck, P. Jansen, L.-M. Meinhardt, P. Ebel, & E. Rukzio, UAM-SUMO: Simulacra of Urban Air Mobility Using SUMO To Study Large-Scale Effects
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Workshops

1. M. Haimerl, P. Jansen, A. Riener and **M. Colley**, Accessible Automated Automotive Workshop Series (A3WS): Accessibility in Mobility
In Proc. MuC EA 2025, Gesellschaft für Informatik e.V., Website: https://a3ws.github.io/MuC2025/
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Extended abstracts / Posters

1. L.-M. Meinhardt*, **M. Colley***, A. Balci, A. Degenhard, P. Jansen, & E. Rukzio, Rethinking Simulator Fidelity in Urban Air Mobility: A Null-Result Investigation into the Effects of Induced Arousal and Motion Cues
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2. M. Ren, P. Zheng, & **M. Colley**, VLM-driven Risk-Adaptive HUD Interactions for Trust Calibration in Automated Driving
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11. M. Walch, D. Lehr, **M. Colley** and M. Weber, Don't You See Them? Towards Gaze-Based Interaction Adaptation for Driver-Vehicle Cooperation
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Videos

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Workshop position papers

1. **M. Colley**, & E. Rukzio, Challenges of Explainability, Cooperation, & External Communication of Automated Vehicles at CHI 2022
2. M. Walch, **M. Colley**, P. Hock, E. Rukzio, & M. Weber, Turn Drivers Into Users and Keep Them Out-Of-The-Loop to Save Energy at CHI 2020
3. **M. Colley**, M. Walch, & E. Rukzio, Towards Reducing Energy Waste through Usage of External Communication of Autonomous Vehicles at CHI 2020